

RWorksheet_prado#3b.Rmd

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```
# Q1a: Create vectors based on the table data
Respondent <- 1:20
Fathers_Occupation_Code <- c(2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 3, 2, 2, 2, 2, 2)
Sex_Code <- c(1, 3, 3, 3, 1, 2, 3, 1, 1, 1, 3, 2, 1, 3, 2, 1, 3, 1, 1, 1) # Note: Assuming the table co
Persons_at_Home <- c(5, 7, 3, 8, 5, 9, 6, 7, 8, 4, 7, 5, 4, 7, 8, 8, 3, 11, 7, 6)
Siblings_at_School <- c(6, 4, 4, 1, 2, 1, 5, 3, 1, 2, 3, 2, 5, 5, 2, 1, 2, 5, 3, 2)
Types_of_Houses_Code <- c(1, 2, 3, 1, 1, 3, 3, 1, 2, 3, 2, 3, 2, 2, 3, 3, 3, 3, 3, 2)

# Create the data frame
df1 <- data.frame(
  Respondent,
  Fathers_Occupation_Code,
  Sex_Code,
  Persons_at_Home,
  Siblings_at_School,
  Types_of_Houses_Code
)

print("Data Frame 1:")
```

```
## [1] "Data Frame 1:"
```

```
df1
```

##	Respondent	Fathers_Occupation_Code	Sex_Code	Persons_at_Home
## 1	1	2	1	5
## 2	2	2	3	7
## 3	3	1	3	3
## 4	4	2	3	8
## 5	5	2	1	5
## 6	6	2	2	9
## 7	7	2	3	6
## 8	8	2	1	7
## 9	9	2	1	8
## 10	10	2	1	4
## 11	11	1	3	7
## 12	12	2	2	5
## 13	13	2	1	4
## 14	14	2	3	7
## 15	15	3	2	8
## 16	16	2	1	8
## 17	17	2	3	3
## 18	18	2	1	11
## 19	19	2	1	7

```
## 20      20      2      1      6
##  Siblings_at_School Types_of_Houses_Code
## 1      6      1
## 2      4      2
## 3      4      3
## 4      1      1
## 5      2      1
## 6      1      3
## 7      5      3
## 8      3      1
## 9      1      2
## 10     2      3
## 11     3      2
## 12     2      3
## 13     5      2
## 14     5      2
## 15     2      3
## 16     1      3
## 17     2      3
## 18     5      3
## 19     3      3
## 20     2      2
```

##Write a R program to create an empty data frame. Using the following codes:

```
df = data.frame(Ints=integer(),
Doubles=double(), Characters=character(),
Logicals=logical(),
Factors=factor(),
stringsAsFactors=FALSE)

print("Structure of the empty dataframe:")
```

```
## [1] "Structure of the empty dataframe:"
```

```
print(str(df))
```

```
## 'data.frame':  0 obs. of  5 variables:
## $ Ints      : int
## $ Doubles   : num
## $ Characters: chr
## $ Logicals  : logi
## $ Factors   : Factor w/ 0 levels:
## NULL
```

Q1b: Get structure of the data

```
print("Structure (str):")
```

```
## [1] "Structure (str):"
```

```
str(df1)
```

```
## 'data.frame':  20 obs. of  6 variables:
## $ Respondent      : int  1 2 3 4 5 6 7 8 9 10 ...
## $ Fathers_Occupation_Code: num  2 2 1 2 2 2 2 2 2 2 ...
## $ Sex_Code        : num  1 3 3 3 1 2 3 1 1 1 ...
## $ Persons_at_Home  : num  5 7 3 8 5 9 6 7 8 4 ...
```

```
## $ Siblings_at_School      : num  6 4 4 1 2 1 5 3 1 2 ...
## $ Types_of_Houses_Code    : num  1 2 3 1 1 3 3 1 2 3 ...

# Get summary of the data
print("\nSummary (summary):")

## [1] "\nSummary (summary):"

summary(df1)

##      Respondent      Fathers_Occupation_Code      Sex_Code      Persons_at_Home
##  Min.       : 1.00      Min.       :1.00           Min.       :1.00      Min.       : 3.0
##  1st Qu.: 5.75      1st Qu.:2.00           1st Qu.:1.00      1st Qu.: 5.0
##  Median :10.50      Median :2.00           Median :1.50      Median : 7.0
##  Mean   :10.50      Mean   :1.95           Mean   :1.85      Mean   : 6.4
##  3rd Qu.:15.25      3rd Qu.:2.00           3rd Qu.:3.00      3rd Qu.: 8.0
##  Max.    :20.00      Max.    :3.00           Max.    :3.00      Max.    :11.0
##  Siblings_at_School Types_of_Houses_Code
##  Min.       :1.00      Min.       :1.0
##  1st Qu.:2.00      1st Qu.:2.0
##  Median :2.50      Median :2.5
##  Mean   :2.95      Mean   :2.3
##  3rd Qu.:4.25      3rd Qu.:3.0
##  Max.    :6.00      Max.    :3.0

# Q1c: Calculate the mean number of siblings attending school
mean_siblings <- mean(df1$Siblings_at_School)

cat("Mean number of siblings attending school:", mean_siblings, "\n")

## Mean number of siblings attending school: 2.95

cat("Is the mean number of siblings attending 5? ", mean_siblings == 5)

## Is the mean number of siblings attending 5? FALSE

# Q1d: Extract the 1st two rows and all the columns
df1[1:2, ]

##      Respondent Fathers_Occupation_Code Sex_Code Persons_at_Home
## 1             1             2             1             5
## 2             2             2             3             7
##  Siblings_at_School Types_of_Houses_Code
## 1                 6             1
## 2                 4             2

# Q1e: Extract 3rd and 5th row with 2nd and 4th column
df1[c(3, 5), c(2, 4)]

##      Fathers_Occupation_Code Persons_at_Home
## 3                 1             3
## 5                 2             5

# Q1f: Select the variable types of houses then store the vector that results as types_houses
types_houses <- df1$Types_of_Houses_Code
print("types_houses vector created:")

## [1] "types_houses vector created:"
```

```
types_houses
```

```
## [1] 1 2 3 1 1 3 3 1 2 3 2 3 2 2 3 3 3 3 2
```

```
# Q1g: Select Males (Sex_Code == 1) whose father occupation was farmer (Fathers_Occupation_Code == 1)  
df1[df1$Sex_Code == 1 & df1$Fathers_Occupation_Code == 1, ]
```

```
## [1] Respondent          Fathers_Occupation_Code Sex_Code  
## [4] Persons_at_Home      Siblings_at_School     Types_of_Houses_Code  
## <0 rows> (or 0-length row.names)
```

```
# Q1h: Select Females (Sex_Code == 2) with Siblings_at_School >= 5  
df1[df1$Sex_Code == 2 & df1$Siblings_at_School >= 5, ]
```

```
## [1] Respondent          Fathers_Occupation_Code Sex_Code  
## [4] Persons_at_Home      Siblings_at_School     Types_of_Houses_Code  
## <0 rows> (or 0-length row.names)
```

```
df = data.frame(Ints=integer(), Doubles=double(), Characters=character(), Logicals=logical(), Factors=factors(),  
print("Structure of the empty dataframe:")
```

```
## [1] "Structure of the empty dataframe:"
```

```
print(str(df))
```

```
## 'data.frame':    0 obs. of  5 variables:  
## $ Ints      : int  
## $ Doubles   : num  
## $ Characters: chr  
## $ Logicals  : logi  
## $ Factors   : Factor w/ 0 levels:  
## NULL
```

```
# Create vectors for the 10-respondent table
```

```
Respondent_Q3 <- 1:10
```

```
Sex_Q3 <- c("Male", "Female", "Female", "Male", "Male", "Female", "Female", "Male", "Female", "Male")
```

```
Fathers_Occupation_Q3 <- c(1, 2, 3, 3, 1, 2, 2, 3, 1, 3) # Codes: 1=Farmer, 2=Driver, 3=Others
```

```
Persons_at_Home_Q3 <- c(5, 7, 3, 8, 6, 4, 4, 2, 11, 6)
```

```
Siblings_at_School_Q3 <- c(2, 3, 0, 5, 2, 3, 1, 2, 6, 2)
```

```
Types_of_Houses_Q3 <- c("Wood", "Congrete", "Congrete", "Wood", "Semi-congrete", "Semi-congrete", "Wood", "Wood", "Wood", "Wood")
```

```
df3 <- data.frame(  
  Respondent = Respondent_Q3,  
  Sex = Sex_Q3,  
  Fathers_Occupation_Code = Fathers_Occupation_Q3,  
  Persons_at_Home = Persons_at_Home_Q3,  
  Siblings_at_School = Siblings_at_School_Q3,  
  Types_of_Houses = Types_of_Houses_Q3  
)
```

```
print("Data Frame 3 (Initial):")
```

```
## [1] "Data Frame 3 (Initial):"
```

```
df3
```

```
##   Respondent  Sex Fathers_Occupation_Code Persons_at_Home Siblings_at_School  
## 1           1  Male                      1                5                2
```

```
## 2      2 Female      2      7      3
## 3      3 Female      3      3      0
## 4      4 Male       3      8      5
## 5      5 Male       1      6      2
## 6      6 Female     2      4      3
## 7      7 Female     2      4      1
## 8      8 Male       3      2      2
## 9      9 Female     1     11      6
## 10     10 Male      3      6      2
```

```
## Types_of_Houses
```

```
## 1      Wood
## 2     Congrete
## 3     Congrete
## 4      Wood
## 5 Semi-concrete
## 6 Semi-concrete
## 7      Wood
## 8 Semi-concrete
## 9 Semi-concrete
## 10     Congrete
```

```
# Q3a: Create a csv file of this and save it as HouseholdData.csv
```

```
write.csv(df3, "HouseholdData.csv", row.names = FALSE)
```

```
cat("File 'HouseholdData.csv' created.\n")
```

```
## File 'HouseholdData.csv' created.
```

```
# Import the csv file into the R environment
```

```
df3_imported <- read.csv("HouseholdData.csv")
```

```
print("Data frame imported:")
```

```
## [1] "Data frame imported:"
```

```
head(df3_imported)
```

```
## Respondent Sex Fathers_Occupation_Code Persons_at_Home Siblings_at_School
## 1      1 Male      1      5      2
## 2      2 Female    2      7      3
## 3      3 Female    3      3      0
## 4      4 Male      3      8      5
## 5      5 Male      1      6      2
## 6      6 Female    2      4      3
```

```
## Types_of_Houses
```

```
## 1      Wood
## 2     Congrete
## 3     Congrete
## 4      Wood
## 5 Semi-concrete
## 6 Semi-concrete
```

```
# Q3b: Convert Sex into factor using factor() and change it into integer
```

```
# Use levels/labels to ensure correct numeric conversion (Male=1, Female=2)
```

```
df3_imported$Sex_Factor <- factor(
  df3_imported$Sex,
  levels = c("Male", "Female"),
  labels = c(1, 2)
```

```

)

df3_imported$Sex_Integer <- as.integer(as.character(df3_imported$Sex_Factor))

print("Data frame with new Sex factors/integers:")

## [1] "Data frame with new Sex factors/integers:"
df3_imported[, c("Sex", "Sex_Integer")]

##      Sex Sex_Integer
## 1   Male           1
## 2 Female           2
## 3 Female           2
## 4   Male           1
## 5   Male           1
## 6 Female           2
## 7 Female           2
## 8   Male           1
## 9 Female           2
## 10  Male           1

# Q3c: Convert Type of Houses into factor and change it into integer
# Define levels/labels for correct numeric conversion

df3_imported$House_Factor <- factor(
  df3_imported$Types_of_Houses,
  levels = c("Wood", "Congrete", "Semi-concrete"),
  labels = c(1, 2, 3)
)

df3_imported$House_Integer <- as.integer(as.character(df3_imported$House_Factor))

print("Data frame with new House factors/integers:")

## [1] "Data frame with new House factors/integers:"
df3_imported[, c("Types_of_Houses", "House_Integer")]

##      Types_of_Houses House_Integer
## 1             Wood           1
## 2          Congrete           2
## 3          Congrete           2
## 4             Wood           1
## 5      Semi-concrete           3
## 6      Semi-concrete           3
## 7             Wood           1
## 8      Semi-concrete           3
## 9      Semi-concrete           3
## 10          Congrete           2

# Q3d: Factor Father's Occupation (Already coded 1, 2, 3 in Fathers_Occupation_Code)
# Convert the existing numeric code into a factor with descriptive labels.

df3_imported$Occupation_Factor <- factor(
  df3_imported$Fathers_Occupation_Code,

```

```

levels = c(1, 2, 3),
labels = c("Farmer", "Driver", "Others")
)

print("Data frame with new Occupation factor:")

```

```
## [1] "Data frame with new Occupation factor:"
```

```
df3_imported[, c("Fathers_Occupation_Code", "Occupation_Factor")]
```

```
##   Fathers_Occupation_Code Occupation_Factor
## 1                      1          Farmer
## 2                      2          Driver
## 3                      3          Others
## 4                      3          Others
## 5                      1          Farmer
## 6                      2          Driver
## 7                      2          Driver
## 8                      3          Others
## 9                      1          Farmer
## 10                     3          Others
```

```
# Q3e: Select Females AND Driver Fathers
```

```
# Use the factor columns for clean logical subsetting
```

```
df3_imported[df3_imported$Sex == "Female" & df3_imported$Occupation_Factor == "Driver", ]
```

```
##   Respondent    Sex Fathers_Occupation_Code Persons_at_Home Siblings_at_School
## 2          2 Female                      2                7                3
## 6          6 Female                      2                4                3
## 7          7 Female                      2                4                1
##   Types_of_Houses Sex_Factor Sex_Integer House_Factor House_Integer
## 2      Congrete      2          2          2          2
## 6   Semi-concrete      2          2          3          3
## 7        Wood        2          2          1          1
##   Occupation_Factor
## 2          Driver
## 6          Driver
## 7          Driver
```

```
# Q3f: Select respondents with Siblings_at_School >= 5
```

```
df3_imported[df3_imported$Siblings_at_School >= 5, ]
```

```
##   Respondent    Sex Fathers_Occupation_Code Persons_at_Home Siblings_at_School
## 4          4   Male                      3                8                5
## 9          9 Female                      1               11                6
##   Types_of_Houses Sex_Factor Sex_Integer House_Factor House_Integer
## 4        Wood        1          1          1          1
## 9   Semi-concrete      2          2          3          3
##   Occupation_Factor
## 4          Others
## 9          Farmer
```

```
#4. Interpret the Graph
```

```
#Overall Trend: The bar chart displays the count of tweets categorized by sentiment (Negative, Neutral,
```

```
#Negative Sentiment: Negative tweets (red bars) generally show the highest count among all sentiment ty
```

```
#Positive Sentiment: Positive tweets (blue bars) are consistently the second highest sentiment category
```

#Neutral Sentiment: Neutral tweets (yellow bars) generally have the lowest count across the observed period.

#Comparison of Days:

#July 15, 2020 had the highest overall volume of tweets, driven primarily by the high negative count.

#July 21, 2020 saw a significant increase in both Positive and Neutral sentiments, with Positive becoming the dominant category.

#July 20, 2020 appears to be the day with the lowest tweet volume across all sentiment categories.